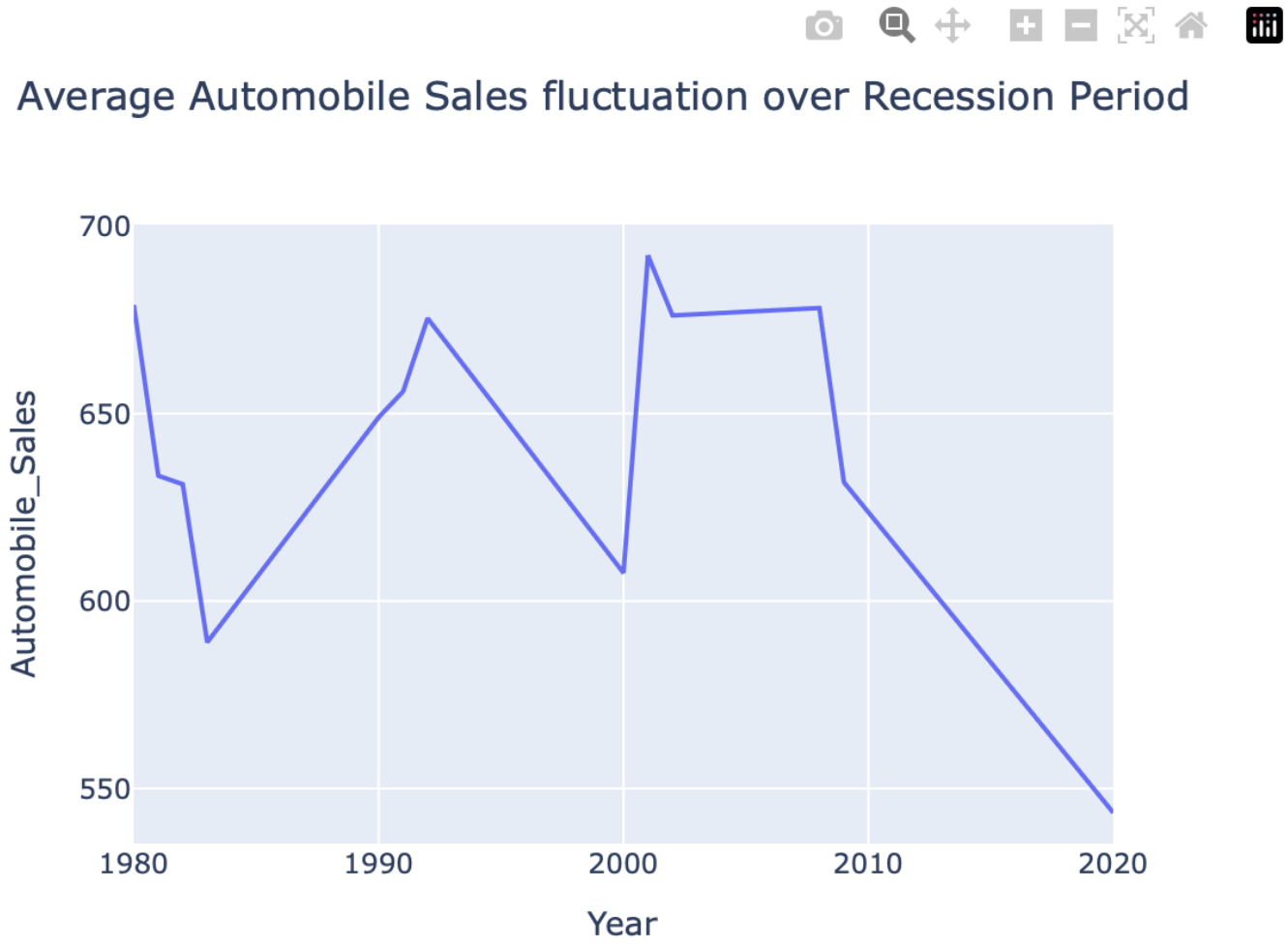
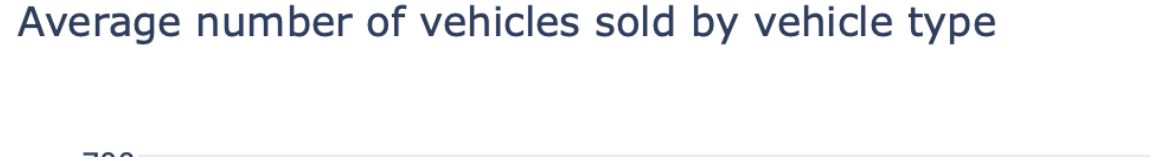
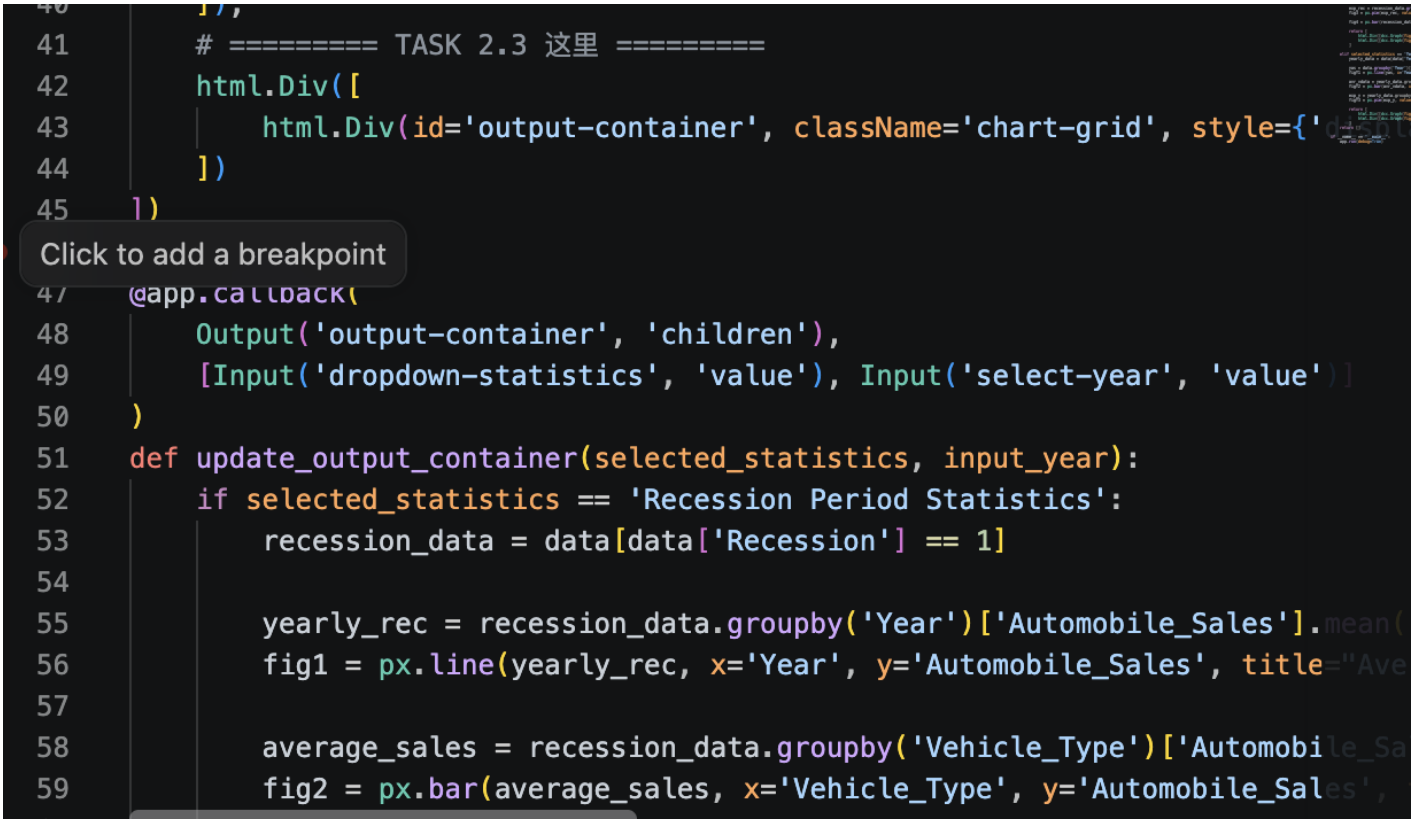
Automobile Sales Statistics Dashboard

Select Statistics:

Recession Period Statistics






41 # ========== TASK 2.3 这里 ==========
42 html.Div([
43 html.Div(id='output-container', className='chart-grid', style={'
44])
45])
Click to add a breakpoint
48 Output('output-container', 'children'),
49 [Input('dropdown-statistics', 'value'), Input('select-year', 'value')]
50)
51 def update_output_container(selected_statistics, input_year):
52 if selected_statistics == 'Recession Period Statistics':
53 recession_data = data[data['Recession'] == 1]
55 yearly_rec = recession_data.groupby('Year')['Automobile_Sales'].mean(
56 fig1 = px.line(yearly_rec, x='Year', y='Automobile_Sales', title="Ave
58 average_sales = recession_data.groupby('Vehicle_Type')['Automobi
59 fig2 = px.bar(average_sales, x='Vehicle_Type', y='Automobile_Sal

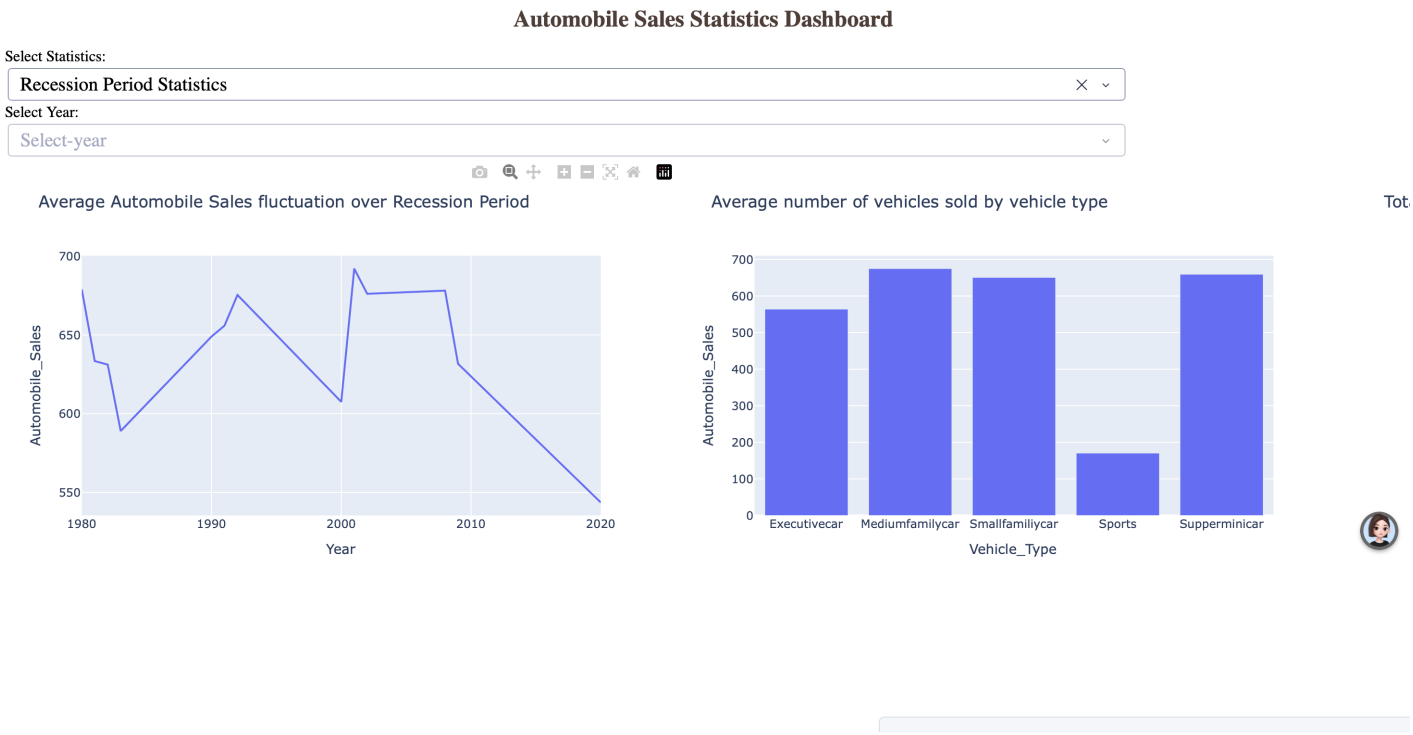
DV0101EN-Final-Assign-Part-2-Questions.py > ...

```python
# ========== Callback 1: 控制年份下拉是否禁用（TASK2.4 第一部分） ==========
@app.callback(
    Output(component_id='select-year', component_property='disabled'),
    Input(component_id='dropdown-statistics', component_property='value')
)
def update_input_container(selected_statistics):
    if selected_statistics == 'Yearly Statistics':
        return False   # 启用下拉，可以选年份
    else:
        return True    # 禁用年份下拉

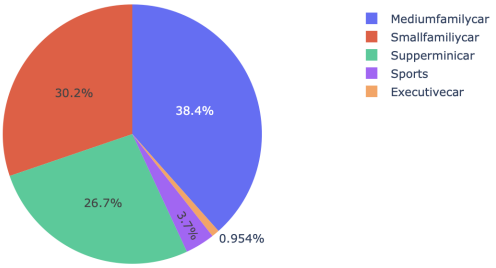
# ========== Callback 2: 绘制图表，输出到 output-container ==========
@app.callback(
    Output(component_id='output-container', component_property='children'),
    [Input(component_id='dropdown-statistics', component_property='value'),
     Input(component_id='select-year', component_property='value')]
)
def update_output_container(selected_statistics, input_year):
    if selected_statistics == 'Recession Period Statistics':
        recession_data = data[data['Recession'] == 1]

        fig1 = px.line(recession_data.groupby('Year')['Automobile_Sales'].mean().
                       x='Year', y='Automobile_Sales',
                       title="Average Automobile Sales fluctuation over Recessio

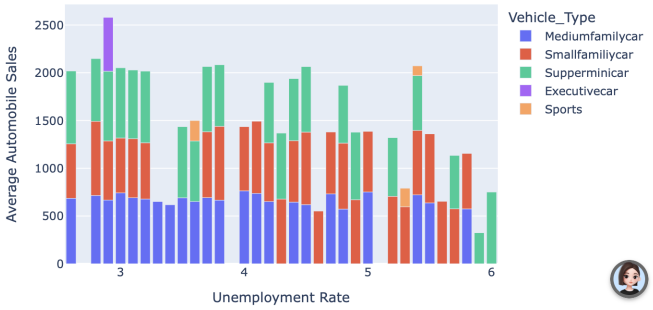
        fig2 = px.bar(recession_data.groupby('Vehicle_Type')['Automobile_Sales'].
```

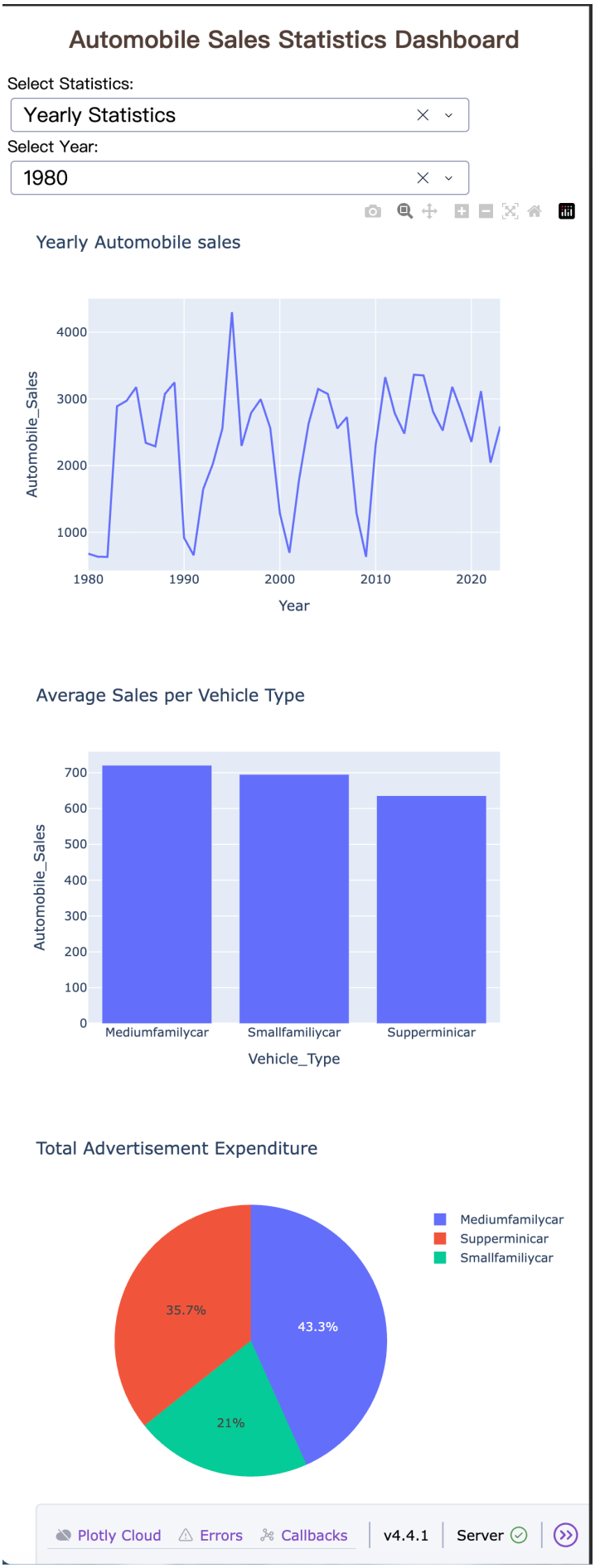

Automobile Sales Statistics Dashboard
Select Statistics:
Recession Period Statistics
Select Year:
Select-year
Average Automobile Sales fluctuation over Recession Period
Automobile_Sales
700
650
600
550
1980
1990
2000
2010
2020
Year
Average number of vehicles sold by vehicle type
Automobile_Sales
700
600
500
400
300
200
100
0
Executivecar
Mediumfamilycar
Smallfamiliycar
Sports
Supperminicar
Vehicle_Type
Tot

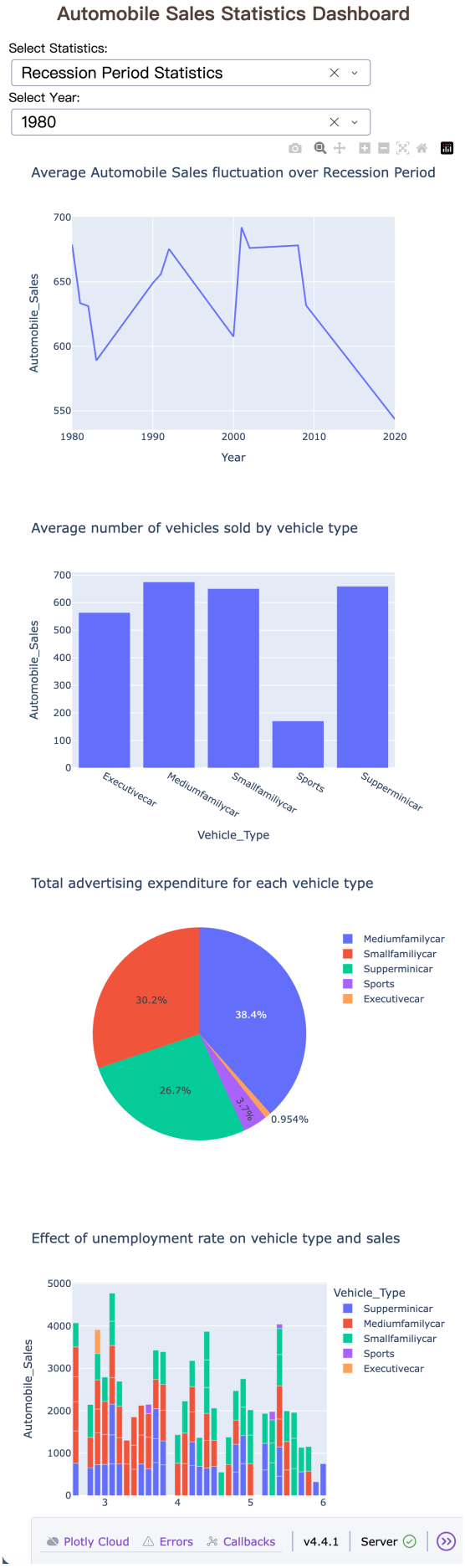
Total advertising expenditure share by vehicle type during recessions

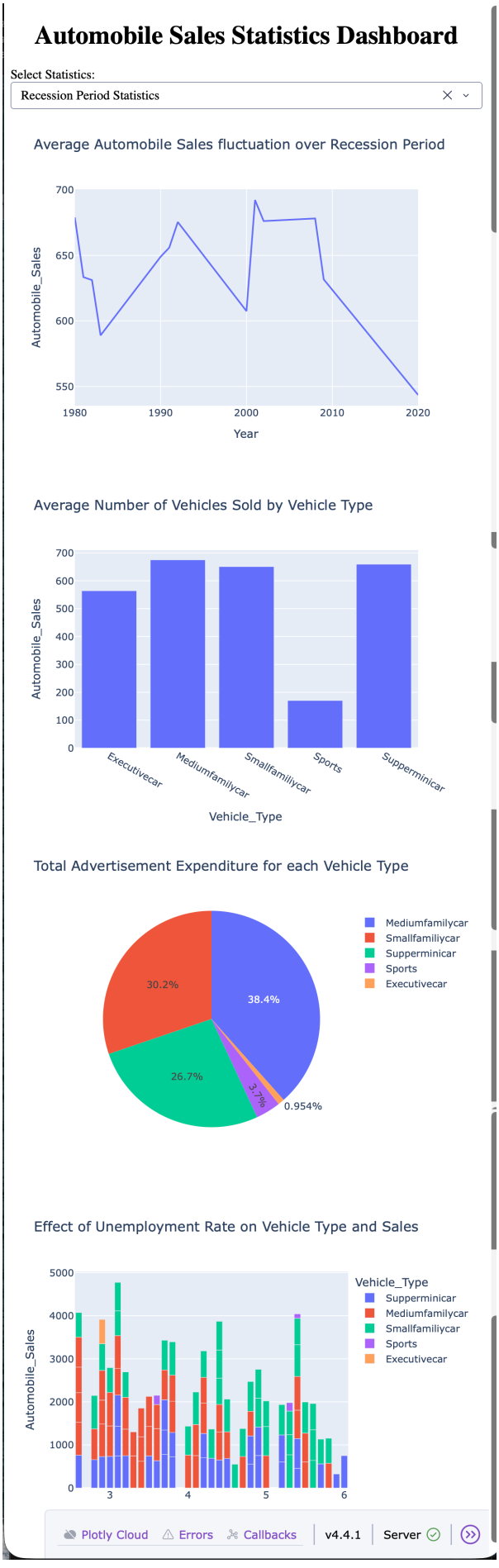


Effect of Unemployment Rate on Vehicle Type and Sales

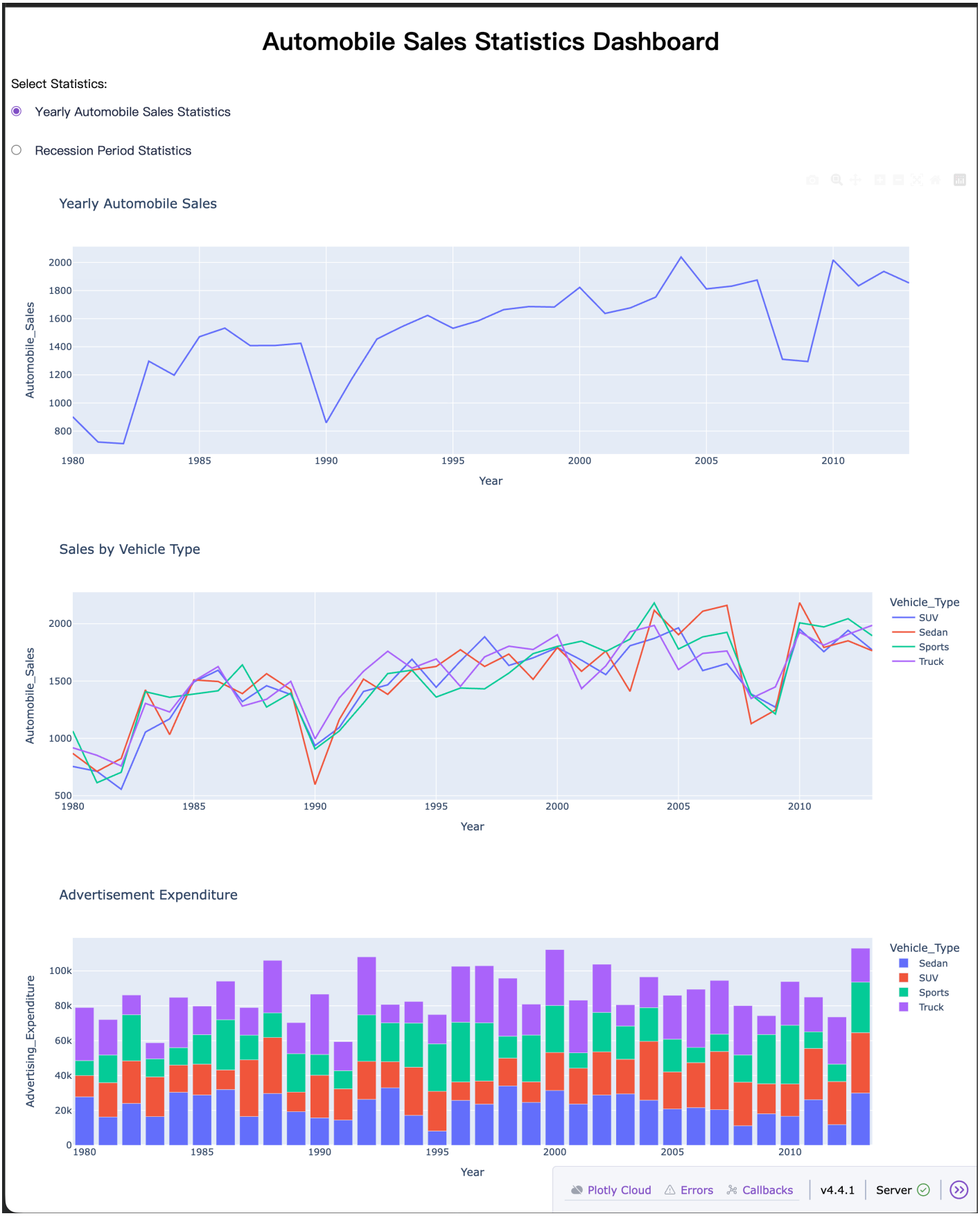



Automobile Sales Statistics Dashboard
Select Statistics:
Yearly Statistics
Select Year:
1980
Yearly Automobile sales
Automobile_Sales
4000
3000
2000
1000
1980
1990
2000
2010
2020
Year
Average Sales per Vehicle Type
Automobile_Sales
700
600
500
400
300
200
100
0
Mediumfamilycar
Smallfamiliycar
Supperminicar
Vehicle_Type
Total Advertisement Expenditure
Mediumfamilycar
Supperminicar
Smallfamiliycar
35.7%
43.3%
21%
Plotly Cloud
Errors
Callbacks
v4.4.1
Server


Automobile Sales Statistics Dashboard
Select Statistics:
Recession Period Statistics
Select Year:
1980
Average Automobile Sales fluctuation over Recession Period
Automobile_Sales
Year
Average number of vehicles sold by vehicle type
Automobile_Sales
Executivecar
Mediumfamilycar
Smallfamiliycar
Sports
Supperminicar
Vehicle_Type
Total advertising expenditure for each vehicle type
Mediumfamilycar
Smallfamiliycar
Supperminicar
Sports
Executivecar
30.2%
38.4%
26.7%
3.7%
0.954%
Effect of unemployment rate on vehicle type and sales
Automobile_Sales
Vehicle_Type
Supperminicar
Mediumfamilycar
Smallfamiliycar
Sports
Executivecar
Plotly Cloud
Errors
Callbacks
v4.4.1
Server



M05FL02_08_dashboard_recession_view

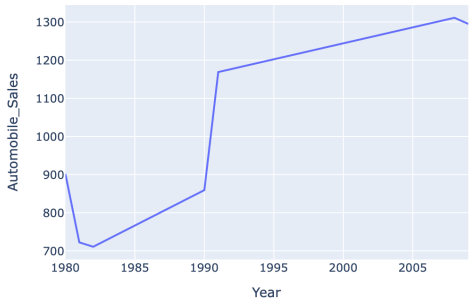

Automobile Sales Statistics Dashboard
Select Statistics:
Yearly Automobile Sales Statistics
Recession Period Statistics
Yearly Automobile Sales
Automobile_Sales
Year
Sales by Vehicle Type
Automobile_Sales
Year
Vehicle_Type
SUV
Sedan
Sports
Truck
Advertisement Expenditure
Advertising_Expenditure
Year
Vehicle_Type
Sedan
SUV
Sports
Truck
Plotly Cloud
Errors
Callbacks
v4.4.1
Server

Automobile Sales Statistics Dashboard

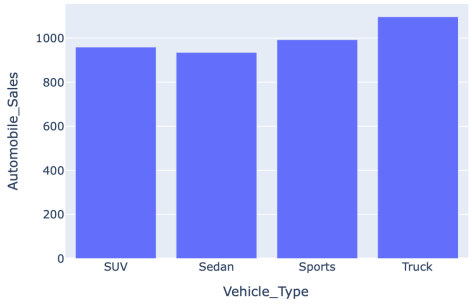
Select Statistics:

- ○ Yearly Automobile Sales Statistics
- • Recession Period Statistics

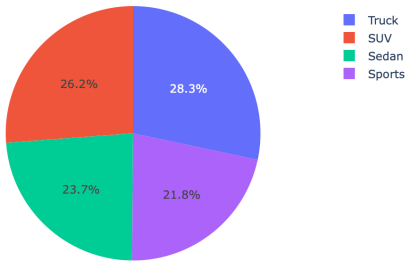
Automobile Sales over Recession Period



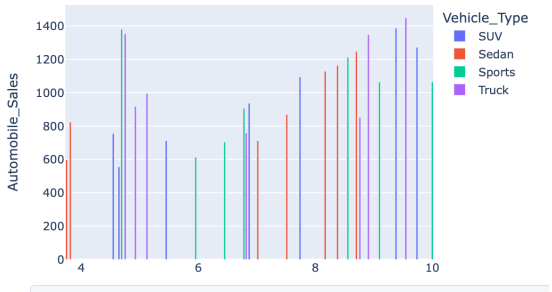
Avg Sales by Vehicle Type (Recession)



Ad Expenditure Share in Recession



Unemployment vs Sales



Plotly Cloud Errors Callbacks v4.4.1 Server